

LINEAR ALGEBRA

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LECTURE 1: THE GEOMETRY OF LINEAR EQUATIONS

A basic problem in linear algebra is solving a SYSTEM OF LINEAR EQUATIONS. We can read the same system in three ways: as separate equations, as a combination of column vectors, or as one matrix equation.

1.1 A two-dimensional system

Consider

$$\begin{aligned}2x - y &= 0, \\ -x + 2y &= 3.\end{aligned}$$

In the ROW PICTURE, each equation describes a line in the xy -plane. The first line passes through the origin and the point $(1, 2)$; the second contains $(-3, 0)$ and $(-1, 1)$. Their intersection is $(1, 2)$ (figure 1), so the simultaneous solution is

$$x = 1, \quad y = 2.$$

Substitution checks both equations. A solution is a point that satisfies every equation at once.

The same system has a different geometric interpretation in the COLUMN PICTURE. Grouping coefficients by unknown gives

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

For vectors c, d and scalars x, y , the sum $xc + yd$ is a LINEAR COMBINATION of c and d . More generally, a linear combination is $a_1c_1 + \cdots + a_kc_k$: multiply each vector by a scalar, then add. We choose the multiples of the two columns that give the right-hand side:

$$1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix},$$

which again yields $x = 1$ and $y = 2$. Geometrically, one copy of the first column followed by two copies of the second column lands exactly on $\mathbf{b} = (0, 3)^T$ (figure 2).

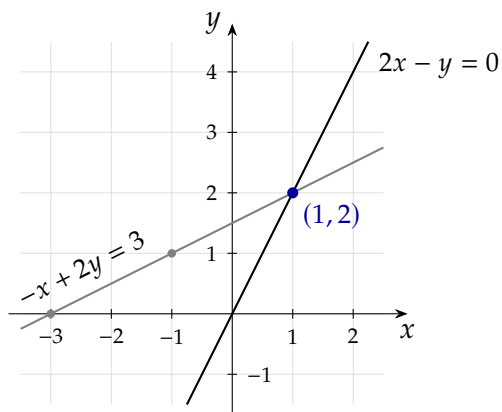


Figure 1: Row picture: the lines $2x - y = 0$ and $-x + 2y = 3$ meet at $(1, 2)$.

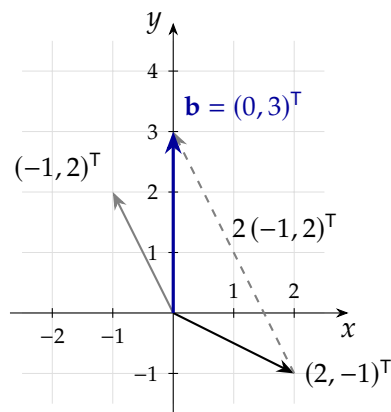


Figure 2: Column picture: one copy of $(2, -1)^T$ plus two copies of $(-1, 2)^T$ gives $\mathbf{b} = (0, 3)^T$.

1.2 The matrix equation

The compact algebraic form is

$$\underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 \\ 3 \end{bmatrix}}_{\mathbf{b}}, \quad A\mathbf{x} = \mathbf{b}.$$

The matrix A is the COEFFICIENT MATRIX, $\mathbf{x}$ is the vector of unknowns, and $\mathbf{b}$ is the right-hand-side vector.

The MATRIX-VECTOR PRODUCT $A\mathbf{x}$ should first be viewed as a combination of the columns of A . If

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

then

$$A\mathbf{x} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

Equivalently, each component is the dot product of a row of A with $\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 2 \cdot 1 + 5 \cdot 2 \\ 1 \cdot 1 + 3 \cdot 2 \end{bmatrix}.$$

The row calculation and column combination are the same multiplication seen from different directions.

1.3 Three equations and three unknowns

For the system

$$\begin{aligned}2x - y &= 0, \\ -x + 2y - z &= -1, \\ -3y + 4z &= 4,\end{aligned}$$

the matrix form is

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

In the row picture, each equation is a plane in three-dimensional space. Two planes generally meet along a line, and a third plane can reduce that common set to one point. That point is the solution. The idea is correct but the geometry quickly becomes difficult to draw in higher dimensions.

The column picture remains algebraically useful. It asks for scalars x, y, z such that

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

Because the right-hand side is the third column, the solution is immediately $\mathbf{x} = (0, 0, 1)^T$. A different right-hand side changes the required coefficients but not the columns being combined.

1.4 Which right-hand sides are reachable?

The equation $A\mathbf{x} = \mathbf{b}$ is solvable precisely when $\mathbf{b}$ is a linear combination of the columns of A . So it is solvable for every $\mathbf{b}$ exactly when combinations of the columns reach every vector in the space containing $\mathbf{b}$.

An n -by- n matrix is INVERTIBLE (OR NONSINGULAR) when every right-hand side has exactly one solution. Geometrically, its columns provide enough independent directions to reach any vector. By contrast, columns are LINEARLY DEPENDENT when one can be formed from the others. In three dimensions, if all three columns lie in one plane, every combination remains in that plane. Right-hand sides outside the plane are unreachable, and the matrix is SINGULAR (figure 3, where $\mathbf{a}_3 = \mathbf{a}_1 + \mathbf{a}_2$, so the three columns span only a plane).

The column picture also works in dimensions we cannot draw: for nine equations and nine unknowns, $A\mathbf{x}$ is still a linear combination of nine columns. The next task is to develop elimination, a systematic procedure for finding the coefficients when a solution exists.

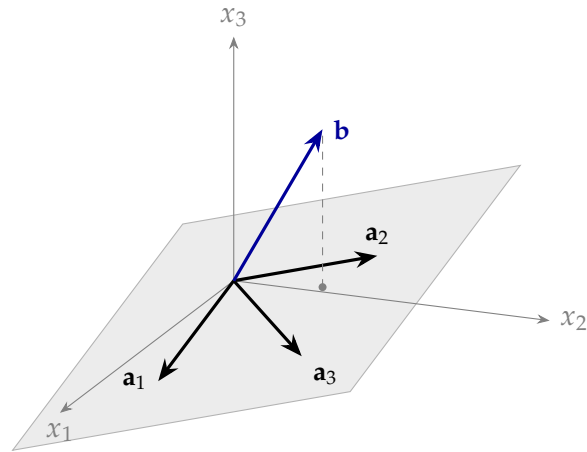


Figure 3: Three dependent columns $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ of a singular 3-by-3 matrix lie in one plane through the origin. Every combination stays in that plane, so a right-hand side $\mathbf{b}$ off the plane cannot be reached.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 1, [The Geometry of Linear Equations](#), lecture by Gilbert Strang.

LECTURE 2: ELIMINATION WITH MATRICES

ELIMINATION simplifies a linear system without changing its solutions. Subtract multiples of rows to create zeros below the diagonal, then solve from the last equation upward. For an invertible matrix, this produces a pivot for every unknown.

2.1 Forward elimination

Consider

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}, \quad A\mathbf{x} = \mathbf{b}.$$

The entry 1 in position (1, 1) is the first **PIVOT**. A pivot is the nonzero entry used to eliminate the entries below it. The corresponding **MULTIPLIER** for row 2 is $3/1 = 3$, so the first row operation is

$$R_2 \leftarrow R_2 - 3R_1.$$

Row 3 already has a zero beneath the first pivot. The matrix becomes

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}.$$

The second pivot is 2. Its multiplier for row 3 is $4/2 = 2$, giving

$$R_3 \leftarrow R_3 - 2R_2$$

and hence

$$A \longrightarrow U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}.$$

The matrix U is **UPPER TRIANGULAR**: its lower left portion is filled with zeros, so every entry below the main diagonal is zero. The pivots are 1, 2, 5.

Every row operation must also be applied to the right-hand side. Appending $\mathbf{b}$ to A creates the **AUGMENTED MATRIX**

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array} \right].$$

The same two operations transform it into

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array} \right],$$

so $A\mathbf{x} = \mathbf{b}$ has become $U\mathbf{x} = \mathbf{c}$, with $\mathbf{c} = (2, 6, -10)^T$.

2.2 Back substitution and pivot failure

The triangular system is

$$\begin{aligned} x + 2y + z &= 2, \\ 2y - 2z &= 6, \\ 5z &= -10. \end{aligned}$$

BACK SUBSTITUTION starts with the last equation and moves upward:

$$z = -2, \quad y = 1, \quad x = 2.$$

Thus $\mathbf{x} = (2, 1, -2)^T$, which also solves the original system.

The complete reduction from A to U is called FORWARD ELIMINATION. A zero cannot serve as a pivot. If the pivot position contains zero but a lower entry in the same column is nonzero, a row exchange moves that entry into the pivot position. If the pivot and every entry below it are zero, no exchange can repair the step. The square matrix is then singular: some right-hand sides have no solution, while others may have infinitely many, so a unique solution for every right-hand side is impossible.

2.3 Elimination as matrix multiplication

Row operations can themselves be represented by matrices. The ELIMINATION MATRIX

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

performs $R_2 \leftarrow R_2 - 3R_1$ when it multiplies on the left:

$$E_{21}A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}.$$

The unchanged rows of E_{21} are rows of the IDENTITY MATRIX

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which satisfies $IA = A$.

The matrix for the next operation is

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

Consequently,

$$E_{32}(E_{21}A) = U.$$

Matrix multiplication is ASSOCIATIVE, so parentheses may move without changing the order of the factors:

$$E_{32}(E_{21}A) = (E_{32}E_{21})A.$$

The product $E = E_{32}E_{21}$ performs both elimination steps. Applying the same matrix to both sides gives

$$E(A\mathbf{x}) = E\mathbf{b} \iff (EA)\mathbf{x} = E\mathbf{b} \iff U\mathbf{x} = \mathbf{c}.$$

The side of multiplication determines what is changed. Left multiplication combines rows, while right multiplication combines columns. Matrix multiplication is generally not commutative: PA and AP need not agree.

2.4 Row exchanges and inverses

A PERMUTATION MATRIX is obtained by rearranging rows of I . For example,

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

exchanges the first two rows of A in PA . Multiplying as AP instead exchanges the first two columns.

An INVERSE of a matrix E is a matrix E^{-1} satisfying $E^{-1}E = I$. Since E_{21} subtracts three times row 1

from row 2, its inverse adds that quantity back:

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{21}^{-1}E_{21} = I.$$

Each row operation can be undone. That is why elimination keeps the same solutions. Applying the inverse operations in reverse order also recovers A from U .

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 2, [Elimination with Matrices](#), lecture by Gilbert Strang.

LECTURE 3: MULTIPLICATION AND INVERSE MATRICES

Matrix multiplication can be understood entry by entry, one column at a time, one row at a time, as a sum of column–row products, or in blocks. Each view makes a different part of the calculation easier to see.

3.1 The row–column rule

Let A be m -by- n and B be n -by- p . Their MATRIX PRODUCT $C = AB$ is the m -by- p matrix whose (i, j) entry is

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Thus the number of columns of A must equal the number of rows of B ; the two outer dimensions determine the shape of the answer.

The ROW–COLUMN RULE defines c_{ij} as the dot product of row i of A and column j of B : multiply corresponding entries, then add. For example,

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 4 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 0 + 0 \cdot 5 & 1 \cdot 1 + 2 \cdot 4 + 0 \cdot (-2) \\ -1 \cdot 2 + 3 \cdot 0 + 1 \cdot 5 & -1 \cdot 1 + 3 \cdot 4 + 1 \cdot (-2) \end{bmatrix} = \begin{bmatrix} 2 & 9 \\ 3 & 9 \end{bmatrix}.$$

3.2 Columns, rows, and outer products

Write the columns of B as $\mathbf{b}_1, \dots, \mathbf{b}_p$. Then

$$AB = [\mathbf{A}\mathbf{b}_1 \quad \cdots \quad \mathbf{A}\mathbf{b}_p].$$

Every column of AB is therefore a linear combination of the columns of A , with coefficients supplied by the corresponding column of B .

The same product can be read by rows. If $\mathbf{a}_i^\top$ is row i of A , then row i of AB is

$$\mathbf{a}_i^\top B.$$

Consequently, every row of AB is a linear combination of the rows of B . The column view explains where the output columns come from; the row view explains the output rows.

Multiplying a column $\mathbf{u}$ by a row $\mathbf{v}^\top$ produces the OUTER PRODUCT $\mathbf{u}\mathbf{v}^\top$. For instance,

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}.$$

All columns of this matrix are multiples of $(2, 3, 4)^\top$, and all rows are multiples of $(1, 6)$.

If $\mathbf{a}_k$ is column k of A and $\mathbf{r}_k^\top$ is row k of B , then

$$AB = \sum_{k=1}^n \mathbf{a}_k \mathbf{r}_k^\top.$$

Matrix multiplication is therefore a sum of outer products, one for each column of A paired with the corresponding row of B .

3.3 Block multiplication

A BLOCK MATRIX is a matrix partitioned into smaller matrices. Blocks may be multiplied by the same row–column rule whenever their dimensions are compatible. For example,

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1B_1 + A_2B_3 & A_1B_2 + A_2B_4 \\ A_3B_1 + A_4B_3 & A_3B_2 + A_4B_4 \end{bmatrix}.$$

The blocks need not all be square or equally sized; only the partitions along each multiplication and addition must match.

3.4 When does an inverse exist?

For a square matrix A , an inverse satisfies

$$A^{-1}A = I = AA^{-1}.$$

An inverse exists exactly when $A\mathbf{x} = \mathbf{b}$ has one solution for every right-hand side. To see how the inverse is built, let $\mathbf{e}_j$ be column j of the identity and $\mathbf{x}_j$ be column j of A^{-1} . Then

$$A\mathbf{x}_j = \mathbf{e}_j.$$

Finding the inverse is therefore the same as solving n systems with the same coefficient matrix and the n columns of I as right-hand sides.

A nonzero solution of $A\mathbf{x} = \mathbf{0}$ proves that A is singular. For example,

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

If an inverse existed, multiplying by it would give $\mathbf{x} = A^{-1}A\mathbf{x} = \mathbf{0}$, contradicting $\mathbf{x} = (3, -1)^T \neq \mathbf{0}$. Geometrically, the two columns lie on the same line (figure 4), so their combinations cannot produce every right-hand side.

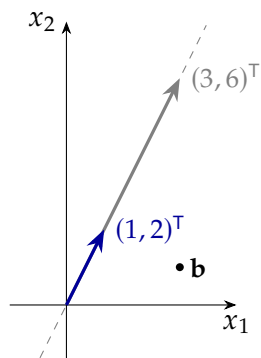


Figure 4: The columns $(1, 2)^T$ and $(3, 6)^T$ of the singular matrix lie on one line through the origin. Every combination stays on that line, so a right-hand side such as $\mathbf{b}$ off the line cannot be reached.

3.5 Gauss–Jordan elimination

GAUSS–JORDAN ELIMINATION computes an inverse by augmenting A with the entire identity matrix and applying row operations until the left half becomes I :

$$\left[A \mid I \right] \longrightarrow \left[I \mid A^{-1} \right].$$

For

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix},$$

the two elimination steps are

$$\begin{aligned} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right] &\xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right] \\ &\xrightarrow{R_1 \leftarrow R_1 - 3R_2} \left[\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$$

Hence

$$A^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}.$$

The block calculation explains why the method works. If E is the product of all row-operation matrices, then

$$E \left[A \mid I \right] = \left[EA \mid E \right].$$

Reaching I on the left means $EA = I$, so $E = A^{-1}$; the right half must therefore be the inverse. If elimination cannot reduce the left half to I , then A is singular and no inverse exists.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 3, [Multiplication and Inverse Matrices](#), lecture by Gilbert Strang.

LECTURE 4: FACTORIZATION INTO $A = LU$

Gaussian elimination lets us write the original matrix as $A = LU$. First we need two rules: taking the inverse or transpose of a product reverses the order of its factors.

4.1 Products, inverses, and transposes

If A and B are invertible, then

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The order reverses because adjacent factors must cancel:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = I.$$

The TRANSPOSE A^T of a matrix is obtained by exchanging its rows and columns: $(A^T)_{ij} = A_{ji}$. Transposing a product also reverses its order:

$$(AB)^T = B^T A^T.$$

For an invertible matrix, we can transpose first or take the inverse first; the result is the same:

$$(A^T)^{-1} = (A^{-1})^T.$$

Indeed, transposing $A^{-1}A = I$ gives $A^T(A^{-1})^T = I$.

4.2 From elimination to $A = LU$

Without row exchanges, elimination gives $EA = U$, where E is the product of the elimination matrices. Multiply by E^{-1} :

$$A = E^{-1}U = LU.$$

This is the LU FACTORIZATION. The matrix $L = E^{-1}$ is LOWER TRIANGULAR: every entry above its diagonal is zero. In this factorization, L has ones on its diagonal and contains the elimination multipliers below it.

For example, eliminating the 8 below the first pivot of

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$$

uses multiplier 4:

$$\underbrace{\begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}}_{E_{21}} \underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_U.$$

The inverse operation adds four times row 1 back to row 2, so

$$A = LU = \underbrace{\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_U.$$

The multiplier appears in L with a positive sign, while the elimination matrix contains its negative.

4.3 Several elimination steps

For a three-by-three matrix, elimination without row exchanges has the form

$$E_{32}E_{31}E_{21}A = U.$$

The inverse-of-a-product rule reverses the order when the factors move to the right:

$$A = \underbrace{E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}}_L U.$$

The reverse order prevents earlier row operations from creating unwanted cross-terms in L . For example, take $E_{31} = I$ and elimination multipliers 2 and 5. The product of elimination matrices is

$$E_{32}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 10 & -5 & 1 \end{bmatrix}.$$

The factor used in $A = LU$, however, is

$$L = E_{21}^{-1}E_{32}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}.$$

Thus, when no row exchanges occur, the multipliers can be copied directly into L while the evolving upper triangular entries are retained in U . There is no need to store the individual elimination matrices.

4.4 Separating the pivots

Sometimes it is useful to split the pivots from U . A **DIAGONAL MATRIX** has zero entries away from its main diagonal. Writing $U = DU'$, where D contains the pivots and U' has ones on its diagonal, gives

$$A = LDU'.$$

For the two-by-two example,

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}.$$

4.5 Cost of elimination

For a dense n -by- n matrix, each stage works on a smaller block in the lower right corner ([figure 5](#)). Clearing the first column touches about n^2 entries, the next stage about $(n-1)^2$, and so on. The main contribution to the work is

$$1^2 + 2^2 + \cdots + n^2 \sim \int_0^n x^2 dx = \frac{1}{3}n^3.$$

Computing L and U costs about $n^3/3$ multiply-subtract steps. Each new right-hand side then takes about n^2 operations for forward and back substitution. Reuse the factors when several systems share A .

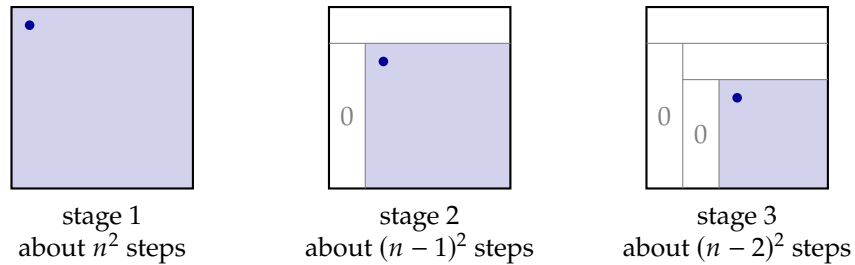


Figure 5: Elimination on a dense n -by- n matrix. Each stage works only on the shaded block, whose top row holds the current pivot (dot); the block shrinks by one row and one column from stage to stage, leaving zeros below the earlier pivots.

4.6 Row exchanges

If a zero appears in a pivot position, a row exchange is required. Left multiplication by

$$P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

swaps rows 1 and 2 of a three-by-three matrix. A permutation matrix is obtained by rearranging the rows of I , so there are $n!$ such matrices of size n -by- n . Their inverses satisfy

$$P^{-1} = P^T.$$

The simple $A = LU$ construction above assumes that no such exchanges are needed; permutation matrices keep track of them when they occur.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 4, [Factorization into \$A = LU\$](#) , lecture by Gilbert Strang.

LECTURE 5: TRANSPOSES, PERMUTATIONS, VECTOR SPACES

We first finish elimination by accounting for row exchanges. Then we study symmetric matrices and vector spaces: collections of vectors that stay closed under addition and scalar multiplication.

5.1 Permutations in elimination

If elimination needs row exchanges, the factorization $A = LU$ becomes

$$PA = LU,$$

where the permutation matrix P puts the rows of A in an order that allows suitable pivots. When no exchanges are needed, $P = I$ and this reduces to $A = LU$.

There are $n!$ permutation matrices of size n -by- n , one for each possible row ordering. Each is invertible, and

$$P^{-1} = P^T, \quad P^T P = I.$$

In exact arithmetic, a row exchange is required when a zero reaches a pivot position. Numerical software may exchange rows even for a nonzero pivot when another available pivot is larger, because division by a very small number can magnify rounding error.

5.2 Transposes and symmetric matrices

Transposing exchanges rows and columns, so an m -by- n matrix becomes an n -by- m matrix. For example,

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}.$$

A square matrix S is a SYMMETRIC MATRIX when

$$S^T = S.$$

Thus its entries mirror across the main diagonal. Symmetric matrices arise naturally from arbitrary rectangular matrices: for every real matrix R , the product $R^T R$ is symmetric. Indeed,

$$(R^T R)^T = R^T (R^T)^T = R^T R.$$

The reverse-order rule for transposes is exactly what makes the product unchanged.

5.3 Vector spaces and closure

A **SCALAR** is a real number used to multiply a vector. A **VECTOR SPACE** is a collection of vectors in which vectors may be added and multiplied by scalars, with the usual algebraic rules, without leaving the collection. Equivalently, every linear combination of vectors in the space must remain in the space.

The set

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^T : x_1, \dots, x_n \in \mathbb{R} \right\}$$

is the fundamental example. Vector addition and scalar multiplication act component by component. A collection is **CLOSED UNDER** an operation when applying that operation to elements of the collection always produces another element of the collection. Thus $\mathbb{R}^n$ is closed under addition and scalar multiplication.

Not every collection of vectors is closed under these operations. The vectors in $\mathbb{R}^2$ whose components are both positive are closed under addition, but not under scalar multiplication: multiplying one of them by -1 leaves the positive quadrant ([figure 6](#)). This collection is therefore not a vector space.

Every vector space contains the zero vector. For any vector $\mathbf{v}$, closure under scalar multiplication requires

$$0\mathbf{v} = \mathbf{0}.$$

The same vector also results from adding $\mathbf{v}$ to its additive inverse.

5.4 Subspaces

A **SUBSPACE** is a vector space contained inside another vector space. It uses the same addition and scalar multiplication. To check a subset of V , verify that it contains $\mathbf{0}$ and stays closed under linear combinations.

If a subspace of $\mathbb{R}^2$ contains a nonzero vector $\mathbf{v}$, it must also contain every multiple $c\mathbf{v}$. These multiples form the entire line through the origin in the direction of $\mathbf{v}$. A line not passing through the origin cannot be a subspace, because it does not contain the zero vector ([figure 7](#)).

The complete possibilities in the first two geometric settings are:

- The subspaces of $\mathbb{R}^2$ are $\mathbb{R}^2$ itself, every line through the origin, and $\{\mathbf{0}\}$.
- The subspaces of $\mathbb{R}^3$ are $\mathbb{R}^3$ itself, every plane through the origin, every line through the origin, and $\{\mathbf{0}\}$.

The smallest case, $\{\mathbf{0}\}$, is the **ZERO SUBSPACE**; the largest subspace of V is V itself.

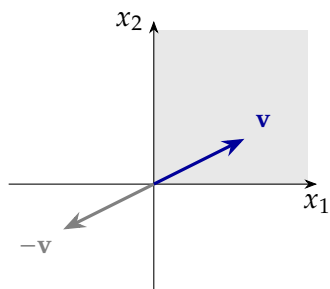


Figure 6: The positive quadrant (shaded) contains $\mathbf{v}$ but not $-\mathbf{v}$, so it is not closed under scalar multiplication.

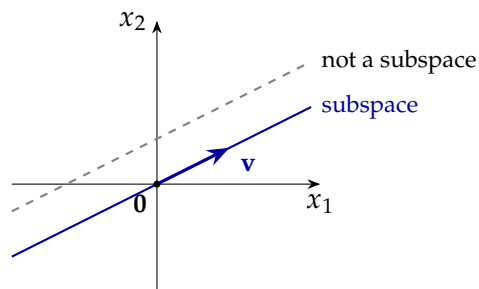


Figure 7: The multiples $c\mathbf{v}$ fill a line through the origin, a subspace of $\mathbb{R}^2$. The parallel dashed line misses $\mathbf{0}$.

5.5 The column space

Let A have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ in $\mathbb{R}^m$. Their COLUMN SPACE is the set of every linear combination of those columns:

$$C(A) = \{x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n : x_1, \dots, x_n \in \mathbb{R}\} = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}.$$

Because a linear combination of linear combinations is still a linear combination of the columns, $C(A)$ is a subspace of $\mathbb{R}^m$.

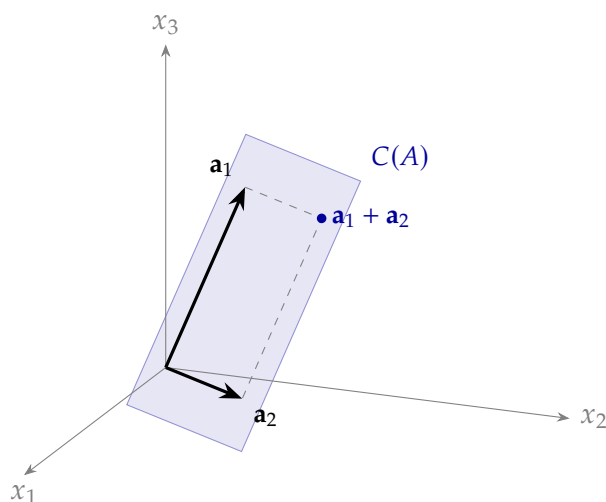


Figure 8: The column space $C(A)$ of the 3-by-2 example is the plane through the origin spanned by $\mathbf{a}_1 = (1, 2, 4)^T$ and $\mathbf{a}_2 = (3, 3, 1)^T$. Every combination, such as $\mathbf{a}_1 + \mathbf{a}_2 = (4, 5, 5)^T$, lies in it, and $A\mathbf{x} = \mathbf{b}$ is solvable only for $\mathbf{b}$ in this plane.

For example, if

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix},$$

then

$$C(A) = \left\{ x_1 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\}.$$

The two columns point in different directions, so their combinations fill a plane through the origin in $\mathbb{R}^3$ ([figure 8](#)). This tells us exactly when $A\mathbf{x} = \mathbf{b}$ can be solved: $\mathbf{b}$ must belong to the column space of A .

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 5, [Transposes, Permutations, Vector Spaces](#), lecture by Gilbert Strang.